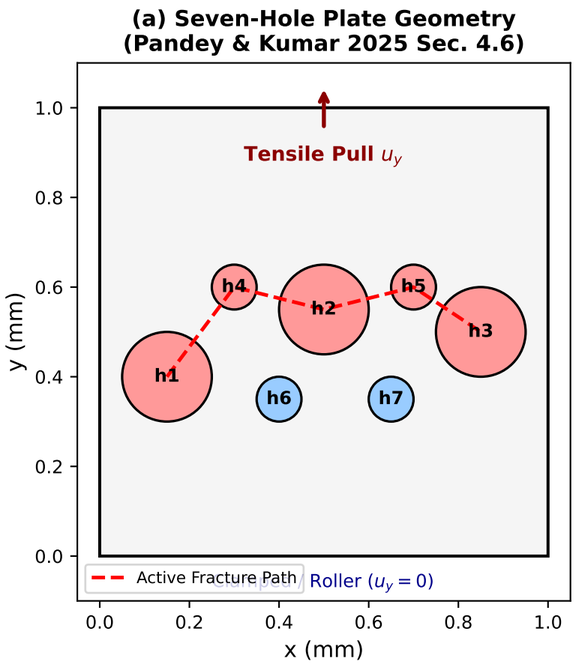
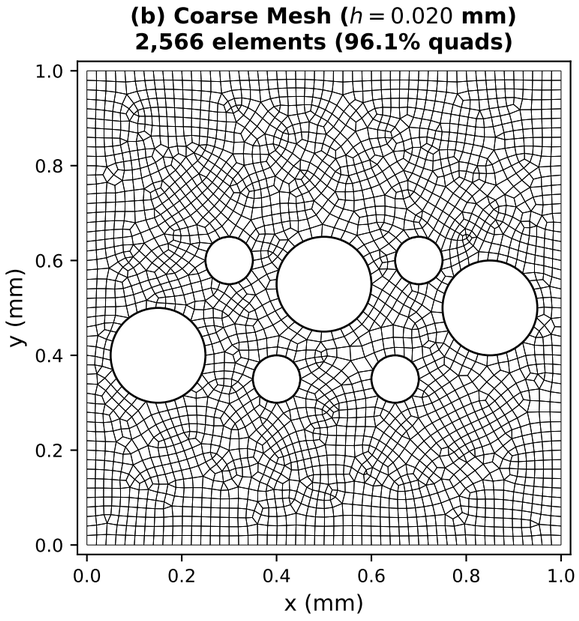
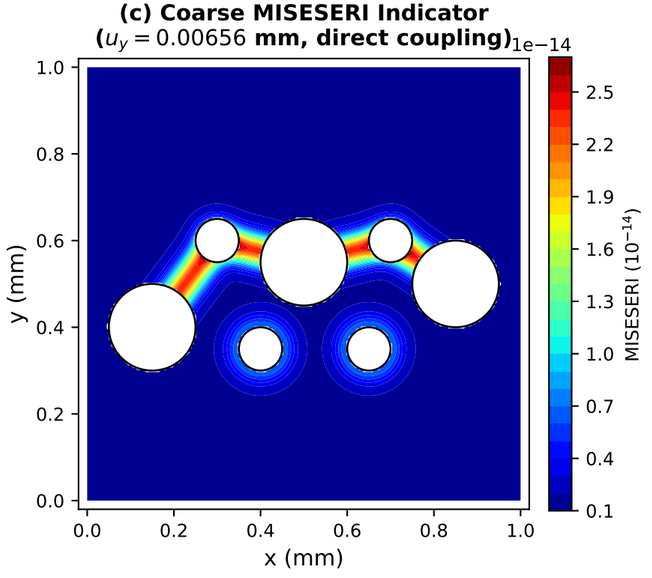
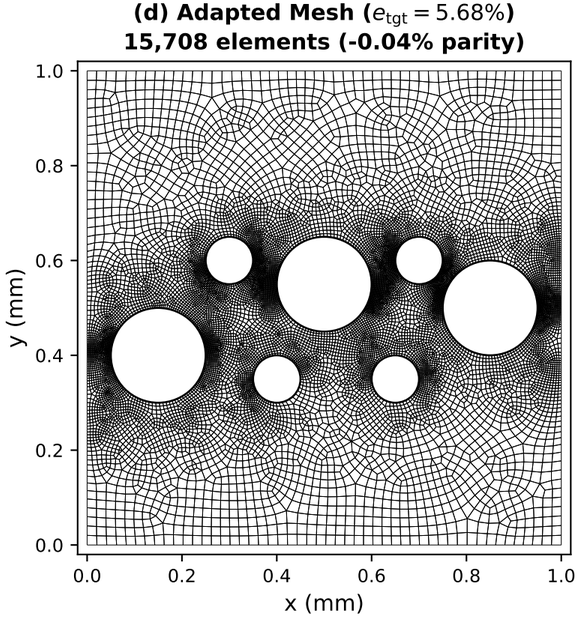

(a) Seven-Hole Plate Geometry
(Pandey & Kumar 2025 Sec. 4.6)
Tensile Pull u_y
h1
h2
h3
h4
h5
h6
h7
Clamped
Active Fracture Path
Roller ($u_y = 0$)
x (mm)
y (mm)


(b) Coarse Mesh ($h = 0.020$ mm)
2,566 elements (96.1% quads)
x (mm)
y (mm)


(c) Coarse MISESERI Indicator
($u_y = 0.00656$ mm, direct coupling)
1e−14
MISESERI (10^{-14})
x (mm)
y (mm)


(d) Adapted Mesh ($e_{tgt} = 5.68\%$)
15,708 elements (-0.04% parity)
x (mm)
y (mm)